

NirogScan Health Report

Patient Information

Name: test1
Age: 24
Gender: Male
Weight: N/A
Height: N/A
Notes: None

Recording Summary

ECG Duration: 95.6s
ECG Sample Rate: 125 Hz
Analysis Window: 72.91s - 77.91s
Window SNR: 25.3 dB
R-peaks Found: 5
ECG Heart Rate: 60.9 ± 0.4 bpm
PPG Duration: 191.1s
PPG Pulse Rate: 90.7 ± 23.2 bpm
SpO2: 81%
Perfusion Index: 206.75%

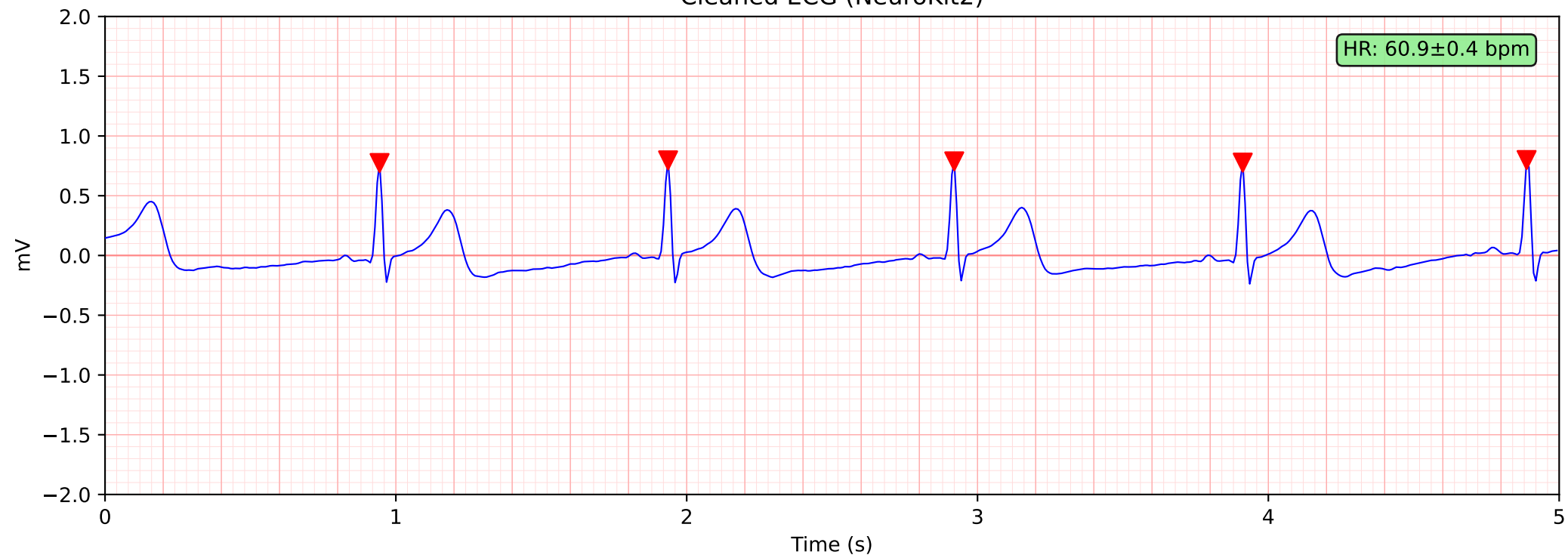
Generated: 2026-01-31 19:43:58

ECG Signal Analysis

Raw ECG - Window: 72.91s-77.91s

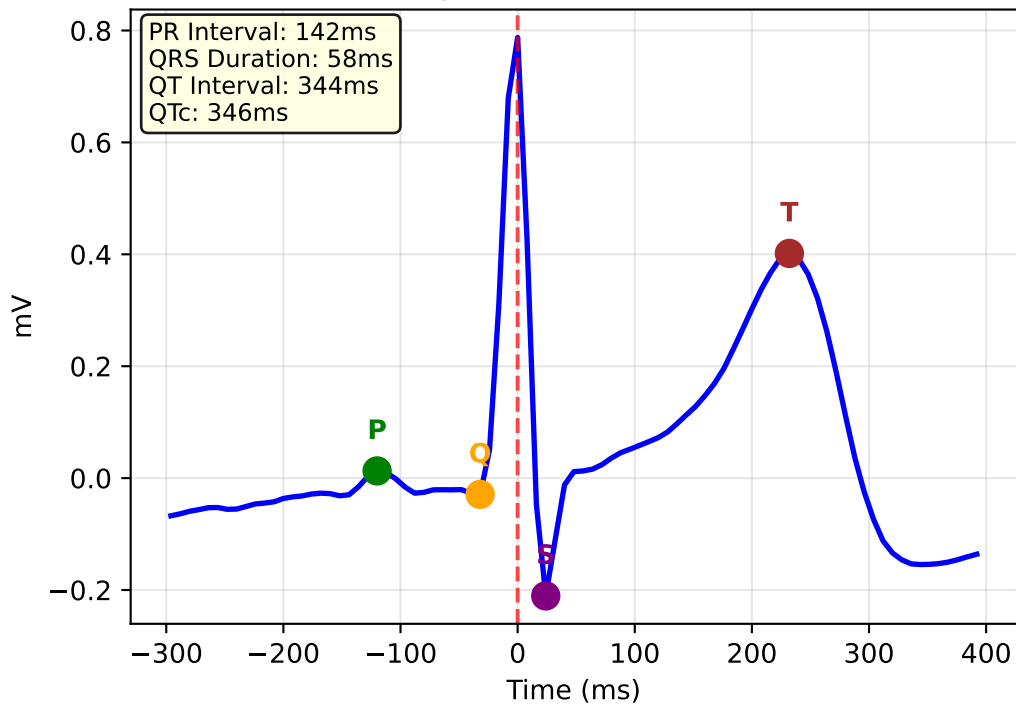


Cleaned ECG (NeuroKit2)

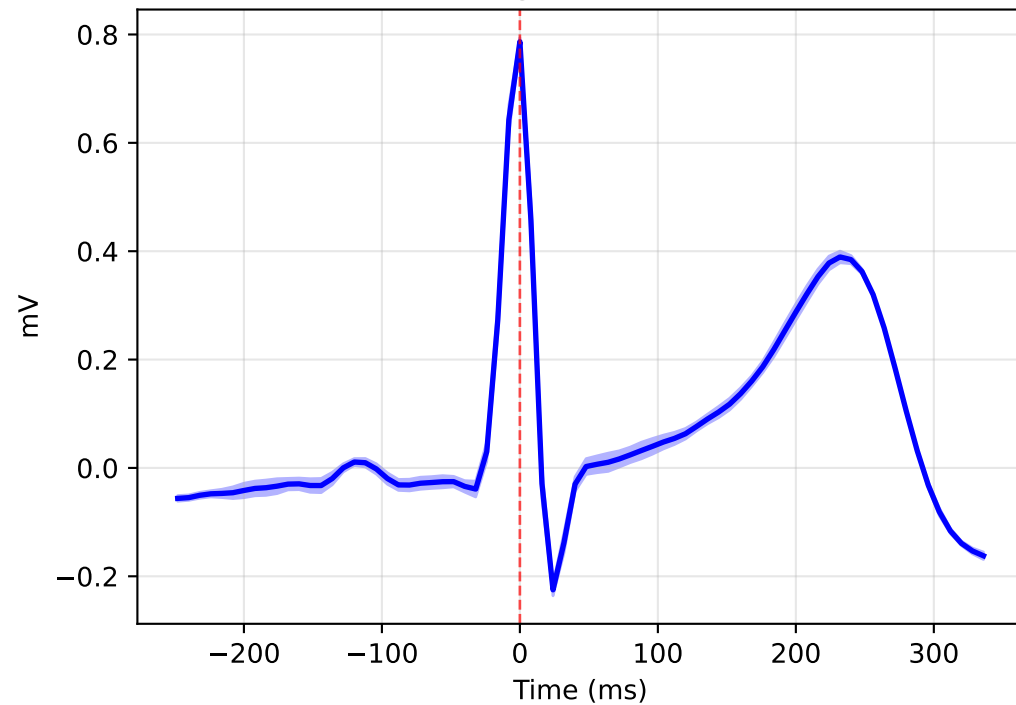


ECG Morphology & Intervals

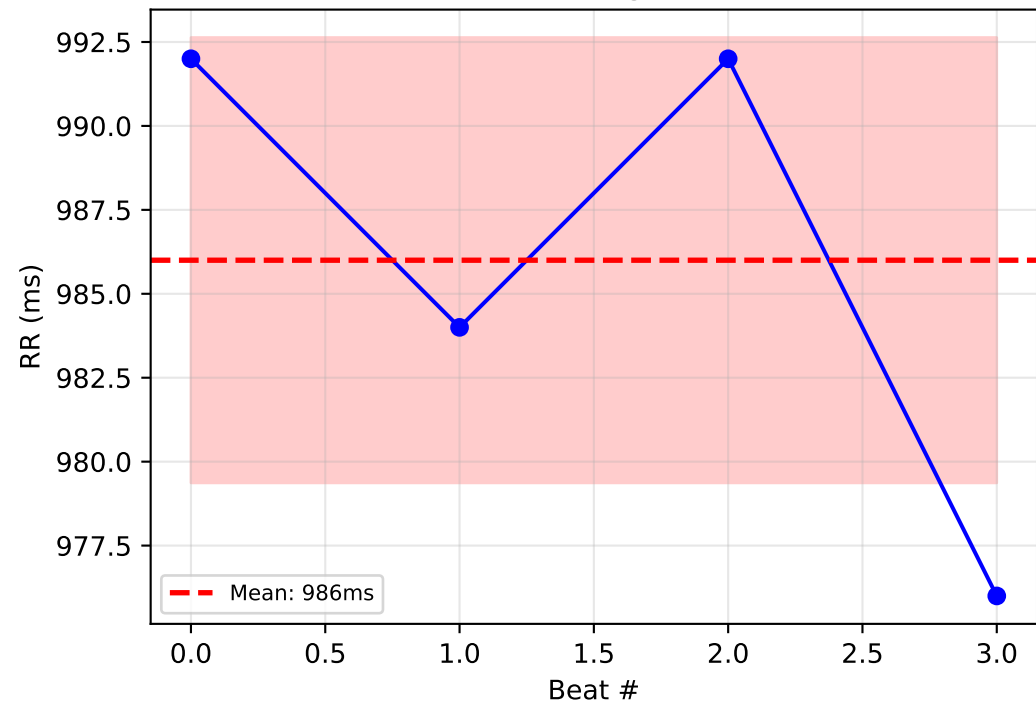
Single Beat with PQRST



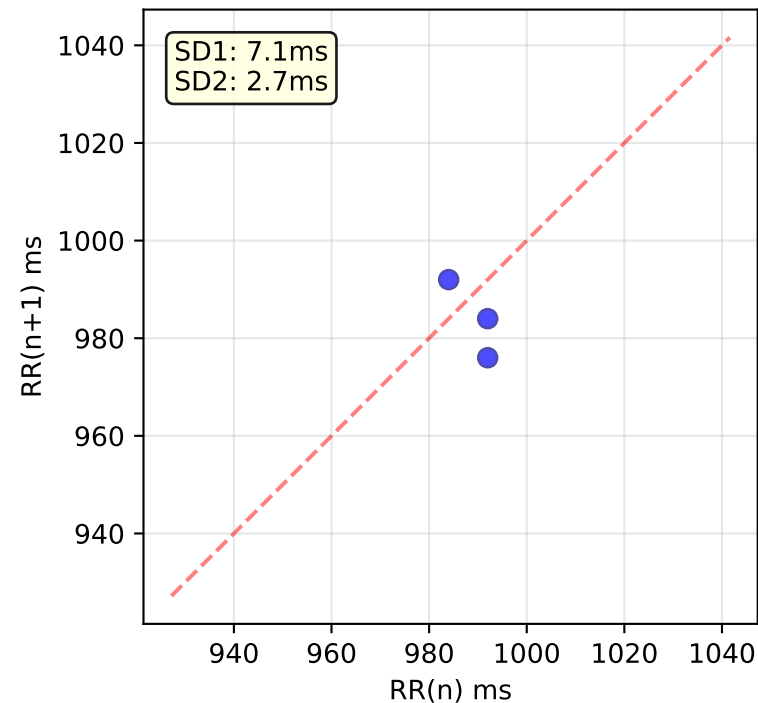
Average Beat (n=3)



RR Tachogram



Poincaré Plot



Heart Rate Variability (HRV) Analysis

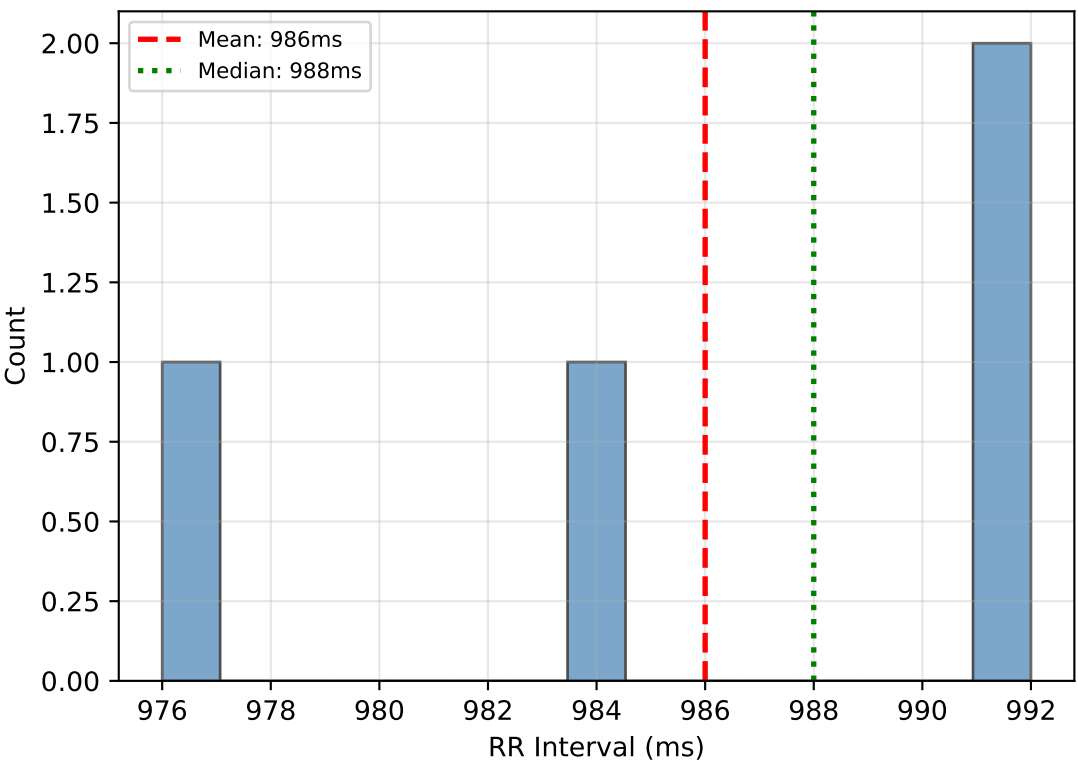
Time Domain

Average RR Interval:	986.00
Heart Rate Variability:	7.66
Short-term HRV:	11.31
Large Beat Changes %:	0.00
Small Beat Changes %:	0.00
Median RR Interval:	988.00

Frequency Domain

Frequency analysis requires
60+ seconds of recording

RR Distribution

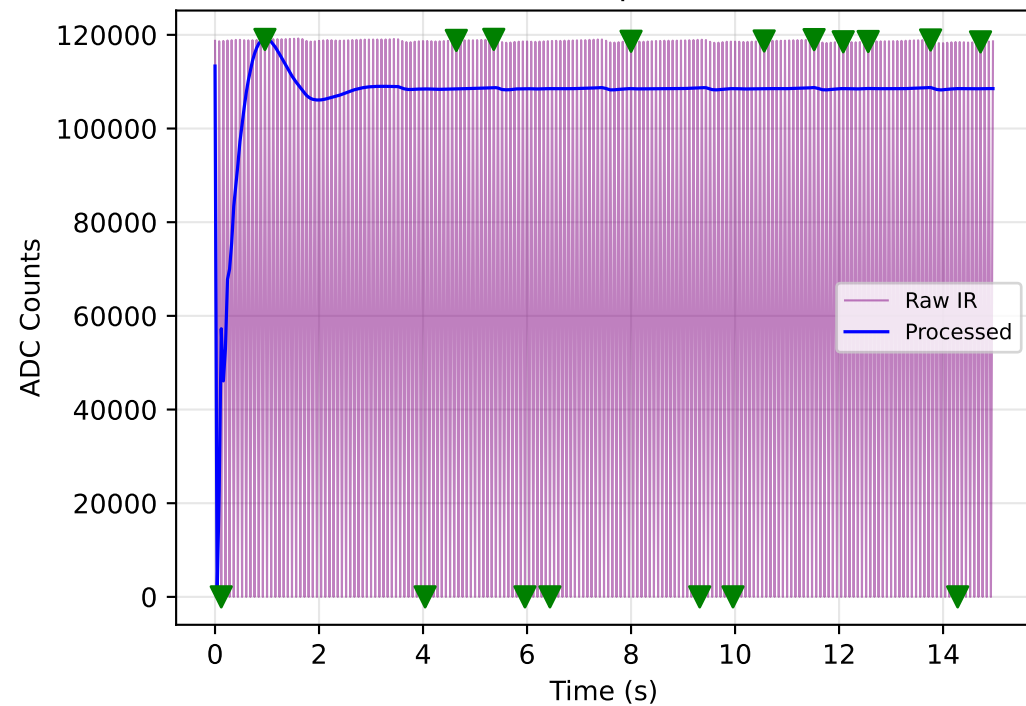


Nonlinear

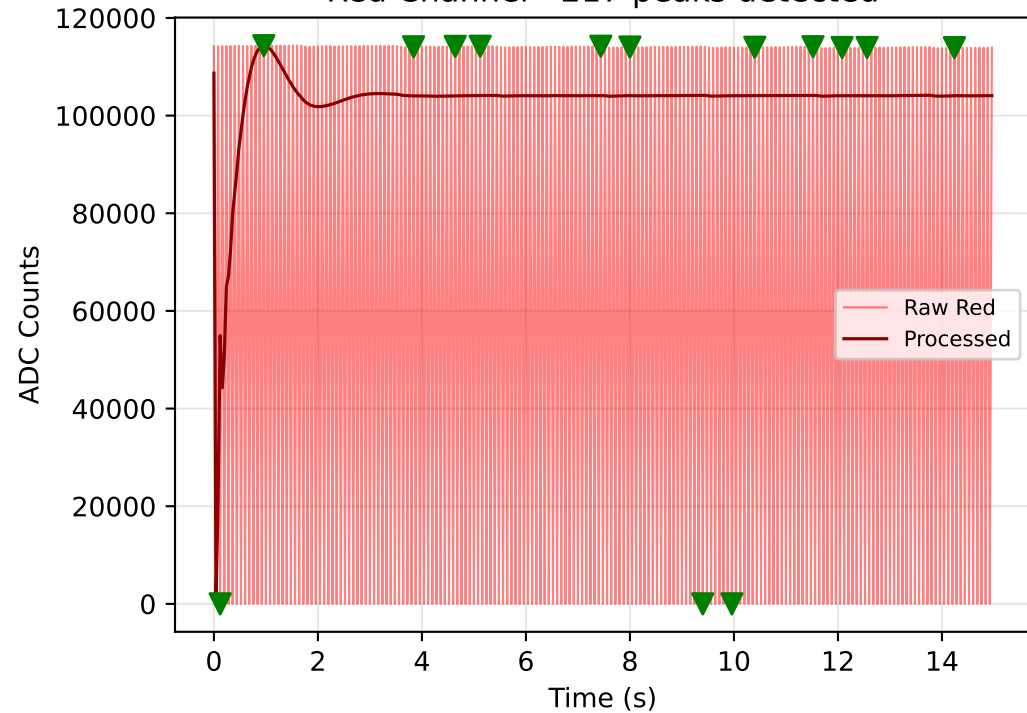
Nonlinear analysis requires
10+ R-peaks

PPG Signal Analysis (MAX30102)

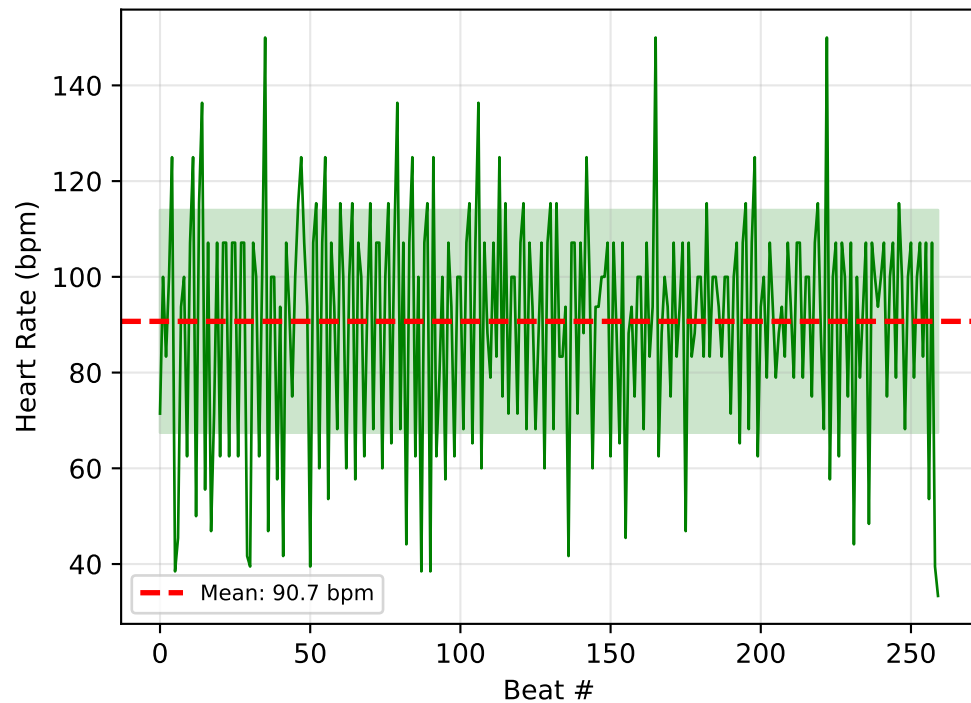
IR Channel - 262 peaks detected



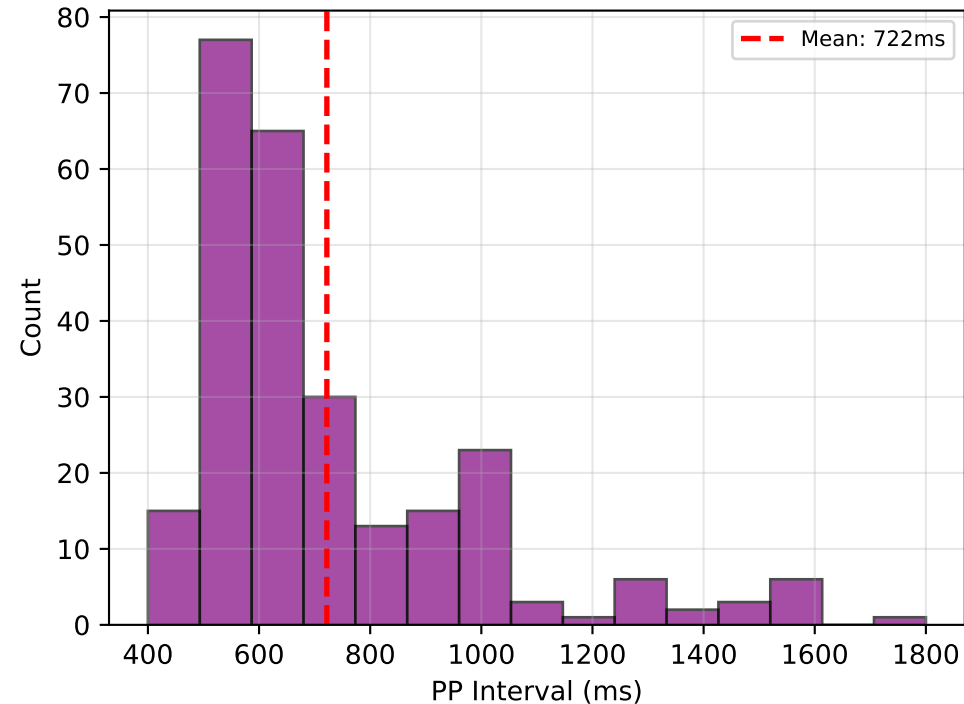
Red Channel - 217 peaks detected



Pulse Rate from PPG



Peak-to-Peak Distribution



PPG Analysis - SpO2 & Pulse Rate Variability

Oxygen Saturation (SpO2)

81%

Oxygen Saturation

R-ratio: 0.998

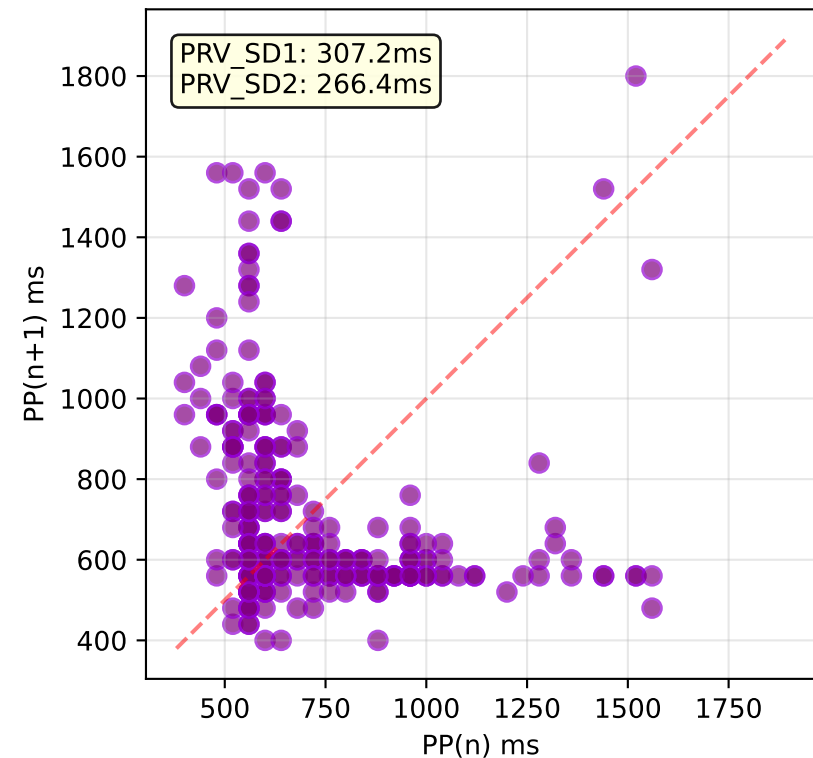
Perfusion & Signal Quality

Perfusion Index (IR):	206.75%
Perfusion Index (Red):	201.49%
Signal Quality:	1.000
Pulse Rate:	90.7 ± 23.2 bpm
PR Range:	33 - 150 bpm
IR Peaks:	262
Red Peaks:	217

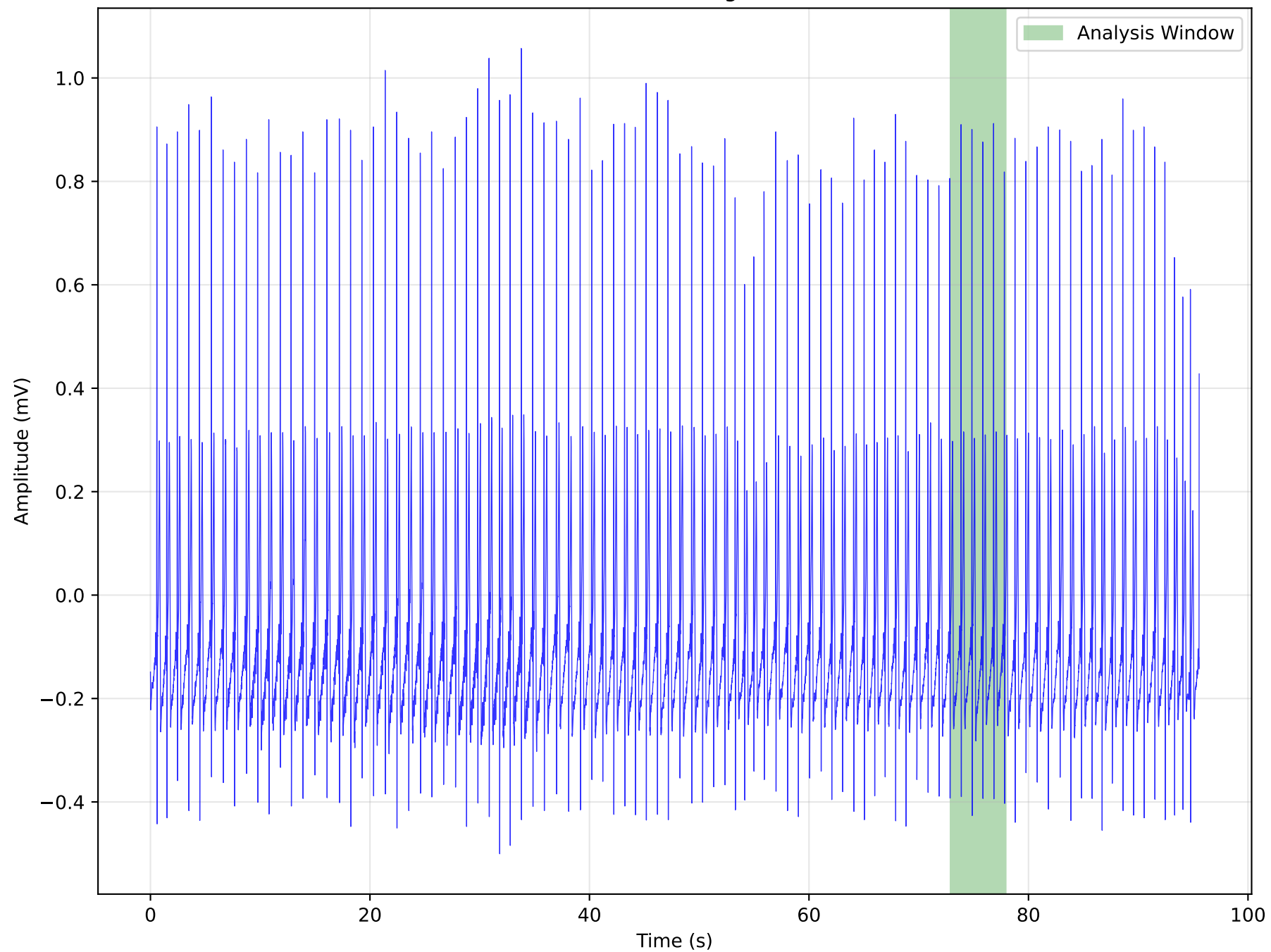
PRV Time Domain

PRV_MeanNN:	731.03
PRV_SDNN:	290.85
PRV_RMSSD:	433.66
PRV_pNN50:	81.99
PRV_pNN20:	93.10

PPG Poincaré Plot



Full ECG Recording (95.6s)



Summary

ECG Results:

Signal SNR:	25.3 dB
Heart Rate:	60.9 ± 0.4 bpm
PR Interval:	142 ms
QRS Duration:	58 ms
QTc:	346 ms
SDNN:	7.7 ms
RMSSD:	11.3 ms

PPG Results:

SpO2:	81%
Perfusion Index:	206.75%
Pulse Rate:	90.7 ± 23.2 bpm
PRV SDNN:	290.8 ms
PRV RMSSD:	433.7 ms

Device:

ECG:	AD8232 @ 125Hz
PPG:	MAX30102 @ 25Hz (100Hz/4 avg)
ADC:	12-bit

Understanding Your Results

(Simple explanations of each measurement)

- **Oxygen Saturation:**
Blood oxygen level. Normal: 95-100%.
- **Perfusion Index:**
Blood flow strength. Higher = better circulation.
- **Pulse Rate:**
Heart rate from blood flow. Should match ECG heart rate.
- **Average RR Interval:**
Average time between heartbeats. Normal: 600-1000ms.
- **Heart Rate Variability:**
How much heartbeat timing varies. Higher = healthier heart. Normal: 50-100ms.
- **Short-term HRV:**
Beat-to-beat changes. Reflects relaxation nervous system. Normal: 20-50ms.
- **Large Beat Changes %:**
Percentage of beats differing >50ms. Higher = good health. Normal: 3-25%.
- **PR Interval:**
Atrial to ventricular time. Normal: 120-200ms.
- **QRS Duration:**
Ventricular contraction time. Normal: 80-120ms.
- **Corrected QT:**
Heart rate adjusted QT. Normal: <440ms.
- **Pulse Rate Variability:**
Variability from PPG. Similar to HRV from ECG.
- **PRV Short-term:**
Beat-to-beat PPG variability.

Note: These are for informational purposes only. Consult a healthcare professional for medical advice.